

Hypothesis: 1/4 of UVA full-time undergraduate students believe a hot dog is a sandwich.

Pre-Analysis:

Data was collected using a Google Form consisting of seven questions exploring opinions on the topic, “Is a hot dog a sandwich?” The responses were downloaded as a CSV file and subsequently cleaned to ensure uniformity and consistency across the dataset. Cleaning steps include isolating the data for undergraduate students and renaming columns. Analysis is completed using Python.

Analysis Methods:

One-sample z-test for proportions to test the hypothesis. The hypothesis makes the claim that there is $p = 0.25$, which is a single population proportion ($H_0: p = p_0$). The alternative hypothesis is two-tailed since we are testing equality ($H_1: p \neq p_0$). The response variable is binary because each student’s response falls into a yes/no category. Yes, a hot dog is a sandwich or no, a hot dog is not a sandwich.

$$np_0 \geq 10 \text{ and } n(1-p_0) \geq 10$$

$$SE = \sqrt{\frac{p_0(1-p_0)}{n}}$$

p_0 : the proportion we are testing against

n = how many people take the survey (sample size)

SE = how much random sampling noise we expect

$$z = \frac{\hat{p} - p_0}{SE}$$

Evaluation of Success:

The output of the analysis is a p-value between 0-1. A result is considered significant if $p < 0.05$, which is a standard metric used in statistical analysis. If $p < 0.05$, results are considered significant and the null hypothesis is rejected. Otherwise, we fail to reject the null hypothesis and do not have sufficient evidence to confirm the hypothesis.